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Supplementary information

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# A generalizable foundation model for analysis of human brain MRI

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# Supplementary Data

## A generalizable foundation model for analysis of human brain MRI

Supplementary DataTable 1. Dataset Image Distribution. In this study we aggregated 34 datasets comprising 48,965 Brain MRI scans from both public sources and private sources such as DFCI/BCH pediatric records, and the RadART study.

| Dataset | Number of Images |
|---|---|
| ABCD | 7197 |
| ADNI | 4825 |
| DFCI/BCH LGG | 4245 |
| OASIS-3 | 4874 |
| MCSA | 2303 |
| SOOP | 3430 |
| ABIDE | 1099 |
| CBTN LGG | 1541 |
| MIRIAD | 523 |
| PPMI | 547 |
| DLBS | 670 |
| RadART LGG | 213 |
| OASIS-2 | 211 |
| DFCI/BCH HGG | 200 |
| QIN-GBM | 99 |
| RIDER | 38 |
| UPENN-GBM | 2520 |
| BRATS23 | 5880 |
| UCSF-PDGM | 990 |
| wu1200 | 1096 |
| LONG579 | 285 |
| BABY | 284 |
| AOMIC | 928 |
| Calgary | 345 |
| HAN | 318 |
| NIMH | 923 |
| ICBM | 809 |
| IXI | 155 |
| NYU | 152 |

| | |
|---|---|
| PING | 738 |
| Pixar | 132 |
| SALD | 181 |
| Petfrog | 268 |
| ABCDV2 | 500 |
| OASIS-1 | 211 |
| ATLAS | 235 |
| Total | 48,965 |

Supplementary Data Table 2. Dataset Link. Summary and Information link for the publicly available datasets used in this study.

| Dataset | Link |
|---|---|
| ABCD | https://nda.nih.gov/abcd |
| ADNI | https://adni.loni.usc.edu/data-samples/ |
| DFCI/BCH LGG | - |
| OASIS-3 | https://sites.wustl.edu/oasisbrains/home/oasis-3/ |
| MCSA | https://www.mayo.edu/research/clinical-trials/cls-20311806 |
| SOOP | https://openneuro.org/datasets/ds004889/versions/1.1.2 |
| ABIDE | https://fcon_1000.projects.nitrc.org/indi/abide/ |
| CBTN LGG | - |
| MIRIAD | https://www.ucl.ac.uk/drc/research-clinical-trials/minimal-interval-resonance-imaging-alzheimers-disease-miriad |
| PPMI | https://www.ppmi-info.org/access-data-specimens/download-data |

| | |
|---|---|
| DLBS | https://fcon_1000.projects.nitrc.org/indi/retro/dlbs.html |
| RadART LGG | - |
| OASIS-2 | https://sites.wustl.edu/oasisbrains/home/oasis-2/ |
| DFCI/BCH HGG | - |
| QIN-GBM | https://wiki.cancerimagingarchive.net/display/Public/QIN+GBM+Treatment+Response |
| RIDER | https://www.cancerimagingarchive.net/collection/rider-neuro-mri/ |
| UPENN-GBM | https://www.cancerimagingarchive.net/collection/upenn-gbm/ |
| BRATS23 | https://www.synapse.org/Synapse:syn51156910/wiki/621282 |
| UCSF-PDGM | https://www.cancerimagingarchive.net/collection/ucsf-pdgm/ |
| wu1200 | https://www.humanconnectome.org/study/hcp-young-adult/document/1200-subjects-data-release |
| LONG579 | - |
| BABY | https://www.sciencedirect.com/science/article/abs/pii/S1053811918302593?via%3Dihub |
| AOMIC | https://www.nature.com/articles/s41597-021-00870-6 |
| Calgary | https://www.sciencedirect.com/science/article/pii/S2352340920301189 |
| HAN | - |
| HIMH | - |
| ICBM | https://ida.loni.usc.edu/collaboration/access/appLicense.jsp |
| IXI | https://brain-development.org/ixi-dataset/ |
| NYU | - |
| PING | https://pmc.ncbi.nlm.nih.gov/articles/PMC4628902/ |
| Pixar | - |
| SALD | https://fcon_1000.projects.nitrc.org/indi/retro/sald.html#:~:text=The%20goals%20of%20the%20SALD,minute%20resting%20state%20fMRI%20scan |
| Petfrog | - |
| ABCDV2 | https://nda.nih.gov/abcd |
| OASIS-1 | https://sites.wustl.edu/oasisbrains/home/oasis-1/ |
| ATLAS | https://fcon_1000.projects.nitrc.org/indi/retro/atlas.html |

Supplementary Data Table 3. Dataset Disease Distribution. The aggregated dataset used for this study encompasses 10 medical settings from healthy to diffuse glioma.

| Condition | Number of Images |
|---|---|
| healthy | 14,981 |
| Alzheimer's | 10,222 |
| PLGG | 5,999 |
| Dementia | 2,749 |
| Stroke | 3,641 |
| Autism | 1,099 |
| Parkinson's | 547 |
| HGG | 200 |
| GBM | 8,537 |
| Diffuse Glioma | 990 |

Supplementary Data Table 4. SSL Pretraining Dataset Distribution. BrainIAC was trained on 32,015 Images across 4 MR sequences (T1w, T2w, T1CE, FLAIR) pooled from 16 datasets.

| Dataset | Number of Images | T1w | T2w | T1CE | FLAIR | Age years (Min- Max) |
|---|---|---|---|---|---|---|
| ABCD | 7197 | 7197 | - | - | | 9-16 |
| ADNI | 4825 | 1844 | - | - | 2981 | 57-58 |
| DFCI/BCH LGG | 4245 | - | 1560 | - | 2685 | 6-20 |
| OASIS-3 | 4874 | 2283 | 1263 | - | 1328 | 42-90 |
| MCSA | 2303 | - | - | 522 | 1781 | 30-89 |
| SOOP | 3430 | 1715 | - | - | 1715 | 35-85 |
| ABIDE | 1099 | 1099 | - | - | - | 7-64 |
| CBTN LGG | 1541 | - | 466 | - | 1075 | 8-21 |
| MIRIAD | 523 | 523 | - | - | - | 65-73 |
| PPMI | 547 | 83 | - | - | 464 | 40-89 |
| DLBS | 670 | 335 | - | - | 335 | 22-88 |
| RadART | 213 | - | - | - | 213 | 8-20 |
| OASIS-2 | 211 | 211 | - | - | - | 60-96 |
| DFCI/BCH HGG | 200 | - | - | - | 200 | 10-21 |
| QIN-GBM | 99 | - | - | 99 | - | 26-81 |
| RIDER | 38 | - | - | 38 | - | 40-80 |

Supplementary Data Table 5. Dataset Sequence Distribution. The aggregated dataset used for this study encompassed 4 sequences T1w, T2w, T1CE, FLAIR.

| Category | Number of Images |
|---|---|
| T1 | 1470 |
| T2 | 1470 |
| T1CE | 1470 |
| FLAIR | 1470 |
| TOTAL | 5880 |

Supplementary Data Table 6. Davies-Bouldin Index clustering scores comparison of the four approaches. KNN clustering (K=4) was performed on the latent features from the backbone of the finetuned models, and Davies-Bouldin Index scores were calculated.

| Data percentage | BrainIAC | MedicalNet | Scratch | BrainSegFounder |
|---|---|---|---|---|
| | | | | |
| 10 | 1.27 | 1.27 | 1.28 | 1.31 |
| 20 | 0.99 | 1.14 | 1.32 | 1.14 |
| 40 | 0.80 | 1.02 | 0.8 | 0.88 |
| 60 | 0.75 | 0.9 | 0.78 | 1.01 |
| 80 | 0.75 | 0.8 | 0.72 | 0.78 |
| 100 | 0.68 | 0.82 | 0.72 | 0.70 |

Supplementary Data Table 7. Brain age Dataset Split. Brain age dataset pool was split into internal (developmental) and external (out of distribution) sets. The internal set was further split into the training-validation and internal (holdout) test set with 80:20 split ratio each. The Internal and external test sets were used for model performance evaluation and comparison.

| DATASET | Min Age (years) | Max Age (Years) |
|---|---|---|
| Internal | | |
| SALD | 19.0 | 35.0 |
| Petfrog | 12.0 | 33.33 |
| PING | 3.0 | 26.7 |
| NYU | 6.47 | 34.9 |
| HAN | 18.0 | 35.0 |
| HIMH | 4.75 | 22.58 |

| | | |
|---|---|---|
| Calgary | 3.0 | 8.0 |
| ICBM | 18.0 | 35.0 |
| AOMIC | 19.0 | 26.0 |
| WU1200 | 22.0 | 35.0 |
| External | | |
| ABCD V2 | 8.92 | 15.42 |
| Pixar | 3.5 | 34.0 |
| Long579 | 5.0 | 9.0 |
| IXI | 3.0 | 34.42 |

Supplementary Data Table 8. Age Distribution of Brain age Dataset. The Internal development set and the external test set consists of subjects with age ranging from 1-40, binned into 4 categories – 0-10Y, 10-20Y, 20-30Y, 30-40Y.

| Age Bin (years) | Number of Images |
|---|---|
| | |
| Internal Set | |
| 0-10 | 1178 |
| 10-20 | 1139 |
| 20-30 | 2224 |
| 30-40 | 636 |
| External Set | |
| 0-10 | 498 |
| 10-20 | 393 |
| 20-30 | 121 |
| 30-40 | 60 |

Supplementary Data Table 9. Brain age prediction performance on the internal test set (n = 1,295) , reported as mean absolute error (MAE) with standard deviation (SD) and standard error of the mean (SE), and 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 3,882) by using the specified percentage of data.

| Data% | MAE | sd | se | 95% CI |
|---|---|---|---|---|
| BrainIAC | | | | |
| 10% | 2.98103668 | 2.4085635 | 0.06693037 | 0.13130393 |
| 20% | 2.60825676 | 2.23417064 | 0.06208425 | 0.12179682 |

| | | | | |
|---|---|---|---|---|
| 40% | 2.37759974 | 2.0756411 | 0.05767895 | 0.11315451 |
| 60% | 2.36327606 | 2.09694228 | 0.05827088 | 0.11431576 |
| 80% | 2.13701737 | 1.90596233 | 0.05296384 | 0.1039044 |
| 100% | 2.26008623 | 1.9873924 | 0.05522666 | 0.10834359 |
| BrainSegFounder | | | | |
| 10% | 4.82682947 | 3.66780564 | 0.10192282 | 0.19995208 |
| 20% | 3.9612175 | 2.95416278 | 0.08209175 | 0.16104752 |
| 40% | 2.86841313 | 2.40952668 | 0.06695713 | 0.13135644 |
| 60% | 2.94126384 | 2.39996545 | 0.06669144 | 0.1308352 |
| 80% | 2.86841313 | 2.40952668 | 0.06695713 | 0.13135644 |
| 100% | 2.89520723 | 2.46316433 | 0.0714377 | 0.27546845 |
| MedicalNet | | | | |
| 10% | 4.25365573 | 3.42934956 | 0.09529648 | 0.18695254 |
| 20% | 3.86507593 | 3.12606146 | 0.08686856 | 0.17041865 |
| 40% | 2.98352317 | 2.60333156 | 0.07234268 | 0.1419218 |
| 60% | 3.24460618 | 2.72288058 | 0.07566477 | 0.14843907 |
| 80% | 2.7903565 | 2.56082404 | 0.07116146 | 0.13960448 |
| 100% | 2.78509073 | 2.51390773 | 0.06985773 | 0.13704682 |
| Scratch | | | | |
| 10% | 3.76407143 | 3.20269744 | 0.08899816 | 0.1745965 |
| 20% | 3.04643822 | 2.412279 | 0.06703362 | 0.13150648 |
| 40% | 3.19825418 | 2.63335974 | 0.07317712 | 0.1435588 |
| 60% | 2.73770528 | 2.40594884 | 0.06685771 | 0.13116139 |
| 80% | 2.62931532 | 2.29554849 | 0.06378985 | 0.12514286 |
| 100% | 2.90725032 | 2.5347644 | 0.0704373 | 0.13818383 |

Supplementary Data Table 10. Brain age prediction performance on the external test set (n = 1,072) , reported as mean absolute error (MAE) with standard deviation (SD) and standard error of the mean (SE), and 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 3,882) by using the specified percentage of data.

| Data% | MAE | sd | se | 95% CI |
|---|---|---|---|---|
| BrainIAC | | | | |
| 10% | 8.09547575 | 3.13129377 | 0.0956371 | 0.18765735 |
| 20% | 6.55058145 | 4.44375003 | 0.13572262 | 0.2663124 |
| 40% | 4.27780239 | 3.45240768 | 0.10544468 | 0.2069016 |
| 60% | 3.85293766 | 2.86054423 | 0.08736777 | 0.17143143 |
| 80% | 4.0784717 | 3.00143486 | 0.09167091 | 0.17987495 |
| 100% | 4.38000155 | 3.6578665 | 0.11171988 | 0.21921468 |

| | | | | |
|---|---|---|---|---|
| BrainSegFounder | | | | |
| 10% | 10.9580224 | 3.2844351 | 0.1003144 | 0.19683506 |
| 20% | 10.0778685 | 3.63573205 | 0.11104384 | 0.21788817 |
| 40% | 9.64217584 | 7.52699743 | 0.22989227 | 0.45109036 |
| 60% | 7.05462764 | 5.5934251 | 0.17083641 | 0.33521203 |
| 80% | 7.25747823 | 3.72995816 | 0.11392173 | 0.2235351 |
| 100% | 5.66202659 | 4.58723662 | 0.14010504 | 0.27491151 |
| MedicalNet | | | | |
| 10% | 8.17749223 | 7.50799464 | 0.22931188 | 0.44995152 |
| 20% | 7.61842219 | 4.36226889 | 0.13323399 | 0.26142927 |
| 40% | 5.29135261 | 4.53746854 | 0.138585 | 0.27192892 |
| 60% | 4.84715563 | 3.24186682 | 0.09901427 | 0.19428396 |
| 80% | 5.96831001 | 3.92051521 | 0.11974179 | 0.23495512 |
| 100% | 5.86159359 | 3.66367827 | 0.11189738 | 0.21956297 |
| Scratch | | | | |
| 10% | 12.3125878 | 5.24250393 | 0.16011845 | 0.31418145 |
| 20% | 7.30439754 | 3.8421646 | 0.11734878 | 0.2302596 |
| 40% | 4.97089008 | 3.7601297 | 0.11484324 | 0.22534327 |
| 60% | 4.61800373 | 3.32020992 | 0.10140705 | 0.19897903 |
| 80% | 4.44153529 | 3.67845471 | 0.11234869 | 0.22044852 |
| 100% | 5.16268037 | 6.54781125 | 0.19998562 | 0.39240807 |

Supplementary Data Table 11. Few-shot brain age prediction performance on the external test set (n = 1,072), reported as mean absolute error (MAE) with standard deviation (SD) and standard error (SE), and 95% CI obtained from 1000 bootstrap samples. Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 4) and K = 5 (n = 20).

| K | MAE | sd | se | 95% CI |
|---|---|---|---|---|
| BrainIAC | | | | |
| K=1 | 9.45478269 | 3.12215539 | 0.095358 | 0.18710969 |
| K=5 | 7.24166801 | 7.40399573 | 0.22613551 | 0.44371891 |
| BrainSegFounder | | | | |
| K=1 | 10.9725042 | 7.52204506 | 0.22974102 | 0.45079356 |
| K=5 | 10.2415207 | 7.53078906 | 0.23000808 | 0.45131759 |
| MedicalNet | | | | |
| K=1 | 17.0488687 | 8.79461918 | 0.24438928 | 0.47944264 |
| K=5 | 15.0280901 | 8.17580881 | 0.22719347 | 0.44570791 |
| Scratch | | | | |
| K=1 | 16.9309073 | 8.81957527 | 0.24508277 | 0.48080313 |

| | | | | |
|---|---|---|---|---|
| K=5 | 14.2854492 | 8.47304646 | 0.23545326 | 0.46191196 |

Supplementary Data Table 12. UCSF-PDGM dataset patient characteristics for IDH mutation classification task.

| | UCSF-PDGM |
|---|---|
| Age : median(range) | 59 (17-94) |
| Sex – M n(%) | 295(59.5%) |
| Sex – F n(%) | 200(40.5%) |
| Vital Status – Dead n(%) | 247(49%) |
| Vital Status – Alive n(%) | 248(51%) |
| IDH-wildtype n(%) | 392 (79.9%) |
| IDH-mutant n(%) | 103 (20.1%) |

Supplementary Data Table 13. IDH mutation prediction performance on the internal test set (n = 99) , reported as AUC with 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 396) by using the specified percentage of data.

| Data% | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | | | |
| 10% | 0.68861454 | 0.56712963 | 0.80661866 |
| 20% | 0.63203018 | 0.49987997 | 0.7688786 |
| 40% | 0.75788752 | 0.5960048 | 0.88271605 |
| 60% | 0.81207133 | 0.68861454 | 0.91426612 |
| 80% | 0.79286694 | 0.6590535 | 0.89368999 |
| 100% | 0.79766804 | 0.648131 | 0.89987997 |
| BrainSegFounder | | | |
| 10% | 0.59259259 | 0.44887689 | 0.71743827 |
| 20% | 0.61162551 | 0.44921982 | 0.75520405 |
| 40% | 0.55898491 | 0.40733882 | 0.71346022 |
| 60% | 0.56001372 | 0.41083676 | 0.70750171 |
| 80% | 0.47856653 | 0.33458505 | 0.62379973 |
| 100% | 0.5941358 | 0.45642147 | 0.72160494 |
| MedicalNet | | | |
| 10% | 0.49657064 | 0.3696845 | 0.62698903 |
| 20% | 0.6351166 | 0.49314129 | 0.75996228 |
| 40% | 0.64026063 | 0.49171811 | 0.76680384 |
| 60% | 0.74622771 | 0.59396433 | 0.88410494 |

| | | | |
|---|---|---|---|
| 80% | 0.7133059 | 0.58089849 | 0.83072702 |
| 100% | 0.68655693 | 0.55006859 | 0.80111454 |
| Scratch | | | |
| 10% | 0.47942387 | 0.34360425 | 0.63170439 |
| 20% | 0.5569273 | 0.39643347 | 0.7297668 |
| 40% | 0.60939643 | 0.46222565 | 0.74557613 |
| 60% | 0.43484225 | 0.27770062 | 0.59333848 |
| 80% | 0.41323731 | 0.26541495 | 0.55149177 |
| 100% | 0.6100823 | 0.49379287 | 0.72223937 |

Supplementary Data Table 14. Few-shot IDH mutation prediction performance on the internal test set (n = 99), reported as AUC with 95% CI obtained from 1000 bootstrap samples. Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 2) and K = 5 (n = 10).

| K | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | | | |
| K=1 | 0.64849108 | 0.49588477 | 0.78532236 |
| K=5 | 0.64746228 | 0.52182785 | 0.76076818 |
| BrainSegFounder | | | |
| K=1 | 0.57750343 | 0.43894033 | 0.70440672 |
| K=5 | 0.6244856 | 0.50474966 | 0.74691358 |
| MedicalNet | | | |
| K=1 | 0.36899863 | 0.24343278 | 0.50008573 |
| K=5 | 0.43758573 | 0.30315501 | 0.57477709 |
| Scratch | | | |
| K=1 | 0.48765432 | 0.34291838 | 0.63516804 |
| K=5 | 0.5569273 | 0.41562071 | 0.68451646 |

Supplementary Data Table 15. Pairwise P values (calculated using DeLong test) comparing IDH-mutation classification AUCs between BrainIAC and each of the three alternative approaches on the internal test set.

| Data Percentage | Scratch | MedicalNet | BrainSegFounder |
|---|---|---|---|
| 10 | 0.03509176 | 0.03161321 | 0.29956658 |
| 20 | 0.50413966 | 0.94277855 | 0.78213977 |
| 40 | 0.18173035 | 0.19299189 | 0.08972368 |
| 60 | 0.00044007 | 0.21471369 | 0.01556231 |

| | | | |
|---|---|---|---|
| 80 | 4.32E-05 | 0.23077667 | 0.00107972 |
| 100 | 0.03890864 | 0.05267224 | 0.03171704 |

Supplementary Data Table16. UPENN-GBM and Brats23/TCGA-GBM dataset patient characteristics.

| | | UPENN-GBM | Brats23/TCGA-GBM |
|---|---|---|---|
| Age: Median (Range) | | 63.0(18.7- 88.5) | 59.0(17.0-84.0) |
| Sex, No | | | |
| | Male | 403 | 83 |
| | Female | 265 | 51 |
| Survival Time in Days, Median,(Range) | | 384(3- 6160) | 383(5-2768) |
| Survival Status | | | |
| | Deceased | 644 | 121 |
| | Alive | 17 | 13 |
| | Lost to Follow-up | 7 | - |

Supplementary Data Table 17. One-year overall survival (binary classification) prediction performance on the internal test set (n = 134) as a function of training-data proportion (10 %–100 %). Metrics include area under the ROC curve (AUC) with 95 % confidence intervals (CI) obtained from 1,000 bootstrap samples, P-values from pairwise DeLong tests versus the BrainIAC model, and concordance index (C-index). All models were fully fine-tuned on the developmental cohort (n = 545) using the varied percentage of data.

| Data% | AUC | 95% CI low | 95% CI high | P-values | C-Index |
|---|---|---|---|---|---|
| BrainIAC | | | | | |
| 10% | 0.62287186 | 0.52529122 | 0.71731071 | - | 0.6018007878 |
| 20% | 0.61648746 | 0.52282706 | 0.71284722 | - | 0.6267867192 |
| 40% | 0.66890681 | 0.57636089 | 0.75674283 | - | 0.6114800225 |
| 60% | 0.70239695 | 0.60640121 | 0.78611111 | - | 0.6619020822 |
| 80% | 0.69422043 | 0.60393145 | 0.77780578 | - | 0.6258863253 |
| 100% | 0.72289427 | 0.63260529 | 0.80243056 | - | 0.640292628 |
| BrainSegFounder | | | | | |
| 10% | 0.54334677 | 0.43871248 | 0.6391297 | 0.28090524 | 0.5329769274 |
| 20% | 0.52671371 | 0.43007672 | 0.63185484 | 0.23294324 | 0.5338773213 |
| 40% | 0.47479839 | 0.37554323 | 0.5781838 | 0.00323589 | 0.479403489 |

| | | | | | |
|---|---|---|---|---|---|
| 60% | 0.53052195 | 0.43332493 | 0.62839382 | 0.00871765 | 0.5232976927 |
| 80% | 0.44119624 | 0.34506888 | 0.54414203 | 5.91E-05 | 0.4556555993 |
| 100% | 0.54760305 | 0.4562808 | 0.6484151 | 0.01139343 | 0.5222847496 |
| MedicalNet | | | | | |
| 10% | 0.36844758 | 0.27281866 | 0.46382728 | 2.08E-05 | 0.59 |
| 20% | 0.43374776 | 0.33411738 | 0.52913306 | 0.00053772 | 0.56 |
| 40% | 0.34610215 | 0.25928539 | 0.4469114 | 3.02E-06 | 0.60 |
| 60% | 0.49288754 | 0.41666387 | 0.57449037 | 0.03822994 | 0.50 |
| 80% | 0.57179659 | 0.47233143 | 0.66644545 | 0.49635612 | 0.43 |
| 100% | 0.53718638 | 0.4421567 | 0.63643593 | 0.0349215 | 0.47 |
| Scratch | | | | | |
| 10% | 0.52161738 | 0.42580925 | 0.62120296 | 0.11908397 | 0.47 |
| 20% | 0.5499552 | 0.4421735 | 0.64248432 | 0.24804971 | 0.47 |
| 40% | 0.3546707 | 0.26276322 | 0.45385865 | 7.24E-05 | 0.60 |
| 60% | 0.42719534 | 0.33422659 | 0.52555164 | 0.00064827 | 0.53 |
| 80% | 0.46152554 | 0.35806452 | 0.55567036 | 0.00091373 | 0.55 |
| 100% | 0.47838262 | 0.37293347 | 0.58210685 | 0.00046681 | 0.54 |

Supplementary Data Table 18. One-year overall survival (binary classification) prediction performance on the external test set (n = 134) as a function of training-data proportion (10 %–100 %). Metrics include area under the ROC curve (AUC) with 95 % confidence intervals (CI) obtained from 1,000 bootstrap samples, P-values from pairwise DeLong tests versus the BrainIAC model, and concordance index (C-index). All models were fully fine-tuned on the developmental cohort (n = 545) using the varied percentage of data.

| Data% | AUC | 95% CI low | 95% CI high | P-values | C-Index |
|---|---|---|---|---|---|
| BrainIAC | | | | | |
| 10% | 0.59255533 | 0.49317572 | 0.68614465 | - | 0.5775561697 |
| 20% | 0.62363067 | 0.52625196 | 0.71563268 | - | 0.5975009011 |
| 40% | 0.65369998 | 0.55331433 | 0.74044825 | - | 0.6038687973 |
| 60% | 0.62351889 | 0.52266376 | 0.71027275 | - | 0.6012255196 |
| 80% | 0.64755198 | 0.55508607 | 0.73307065 | - | 0.6305418719 |
| 100% | 0.66409569 | 0.56336351 | 0.75229153 | - | 0.6309023189 |
| BrainSegFounder | | | | | |
| 10% | 0.49340487 | 0.39558462 | 0.59737033 | 0.17441438 | 0.43 |
| 20% | 0.49144869 | 0.38931087 | 0.5907808 | 0.05820777 | 0.43 |
| 40% | 0.50816007 | 0.41426056 | 0.60552761 | 0.03571261 | 0.42 |
| 60% | 0.4428795 | 0.348055 | 0.53884138 | 0.00692688 | 0.43 |
| 80% | 0.51581712 | 0.41582271 | 0.61605466 | 0.05650232 | 0.43 |
| 100% | 0.5165996 | 0.41211715 | 0.61927398 | 0.03847279 | 0.44 |

| | | | | | |
|---|---|---|---|---|---|
| MedicalNet | | | | | |
| 10% | 0.4361167 | 0.33869048 | 0.53157556 | 0.02516299 | 0.59 |
| 20% | 0.48049408 | 0.38709759 | 0.5903197 | 0.03438644 | 0.60 |
| 40% | 0.39442209 | 0.29735915 | 0.48350939 | 0.0011278 | 0.64 |
| 60% | 0.44768612 | 0.39020512 | 0.50704505 | 0.01378353 | 0.57 |
| 80% | 0.47529622 | 0.37659289 | 0.5715655 | 0.0355549 | 0.57 |
| 100% | 0.48848647 | 0.39335178 | 0.58976079 | 0.0094383 | 0.58 |
| Scratch | | | | | |
| 10% | 0.50011178 | 0.39917002 | 0.59814722 | 0.12465136 | 0.57 |
| 20% | 0.51346971 | 0.41090711 | 0.61089593 | 0.07264076 | 0.57 |
| 40% | 0.37435725 | 0.27643081 | 0.47104851 | 0.00041799 | 0.64 |
| 60% | 0.41001565 | 0.31432763 | 0.50884194 | 0.00410408 | 0.58 |
| 80% | 0.41856696 | 0.31621674 | 0.51354237 | 0.0045636 | 0.63 |
| 100% | 0.49211938 | 0.39107981 | 0.5917505 | 0.01142068 | 0.57 |

Supplementary Data Table 19. Few-shot One-year overall survival (binary classification) prediction performance on the external test set (n = 134), reported as AUC with 95% CI obtained from 1000 bootstrap samples. Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 2) and K = 5 (n = 10).

| K | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | | | |
| K=1 | 0.59296595 | 0.49395161 | 0.68818324 |
| K=5 | 0.60965502 | 0.51812836 | 0.70949261 |
| BrainSegFounder | | | |
| K=1 | 0.42450717 | 0.32704973 | 0.52666331 |
| K=5 | 0.58109319 | 0.47678091 | 0.67958669 |
| MedicalNet | | | |
| K=1 | 0.38272849 | 0.29287354 | 0.47371192 |
| K=5 | 0.55975582 | 0.44757784 | 0.65883177 |
| Scratch | | | |
| K=1 | 0.37102375 | 0.27885585 | 0.46125952 |
| K=5 | 0.48784722 | 0.39547211 | 0.58652554 |

Supplementary Data Table 20. Mild Cognitive Impairment (MCI) prediction performance on the Internal test set (n = 40) as a function of training-data proportion (10 %–100 %). Metrics include area under the ROC curve (AUC) with 95 % confidence intervals (CI) obtained from 1,000 bootstrap samples, P-values from pairwise DeLong tests versus the BrainIAC model, and concordance index (C-index). All models were fully fine-tuned on the developmental cohort (n = 195) using the varied percentage of data.

| Data% | AUC | 95% CI low | 95% CI high | P-values |
|---|---|---|---|---|
| BrainIAC | | | | |
| 10% | 0.70219436 | 0.50148903 | 0.87468652 | - |
| 20% | 0.75862069 | 0.58926332 | 0.88409091 | - |
| 40% | 0.77115987 | 0.60815047 | 0.90603448 | - |
| 60% | 0.78369906 | 0.63942006 | 0.90916928 | - |
| 80% | 0.830721 | 0.68965517 | 0.93730408 | - |
| 100% | 0.88401254 | 0.75854232 | 0.97492163 | - |
| BrainSegFounder | | | | |
| 10% | 0.52978056 | 0.32915361 | 0.7210815 | 0.07141184 |
| 20% | 0.48902821 | 0.29467085 | 0.69592476 | 0.00692641 |
| 40% | 0.53291536 | 0.29459248 | 0.73667712 | 0.10054893 |
| 60% | 0.5799373 | 0.39184953 | 0.74929467 | 0.03662503 |
| 80% | 0.5830721 | 0.39490596 | 0.74929467 | 0.0007927 |
| 100% | 0.62068966 | 0.39811912 | 0.82766458 | 0.31987754 |
| MedicalNet | | | | |
| 10% | 0.56426332 | 0.35109718 | 0.7523511 | 0.43757649 |
| 20% | 0.5830721 | 0.39498433 | 0.76802508 | 0.19551504 |
| 40% | 0.65673981 | 0.47335423 | 0.81191223 | 0.21418523 |
| 60% | 0.74451411 | 0.54851097 | 0.89976489 | 0.6785794 |
| 80% | 0.77429467 | 0.61755486 | 0.90289969 | 0.46221179 |
| 100% | 0.82131661 | 0.67084639 | 0.94051724 | 0.44389105 |
| Scratch | | | | |
| 10% | 0.54231975 | 0.34482759 | 0.73989028 | 0.2668528 |
| 20% | 0.70219436 | 0.48902821 | 0.87468652 | 0.68401447 |
| 40% | 0.62068966 | 0.41998433 | 0.81191223 | 0.24219634 |
| 60% | 0.49211938 | 0.39107981 | 0.5917505 | 0.20946119 |
| 80% | 0.79623824 | 0.64255486 | 0.92170846 | 0.64570432 |
| 100% | 0.77429467 | 0.55172414 | 0.93746082 | 0.91228092 |

Supplementary Data Table 21. Few-shot Mild Cognitive Impairment (MCI) prediction performance on the internal test set (n = 40), reported as AUC with 95% CI obtained from 1000

bootstrap samples. Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 2) and K = 5 (n = 10).

| K | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | | | |
| K=1 | 0.69592476 | 0.51410658 | 0.84326019 |
| K=5 | 0.79937304 | 0.64890282 | 0.92476489 |
| BrainSegFounder | | | |
| K=1 | 0.60501567 | 0.41065831 | 0.79623824 |
| K=5 | 0.53291536 | 0.3354232 | 0.71786834 |
| MedicalNet | | | |
| K=1 | 0.55799373 | 0.36050157 | 0.74302508 |
| K=5 | 0.63636364 | 0.45133229 | 0.79623824 |
| Scratch | | | |
| K=1 | 0.49529781 | 0.29467085 | 0.69278997 |
| K=5 | 0.51724138 | 0.29145768 | 0.7273511 |

Supplementary Data Table 22. Time to stroke prediction performance on the internal test set (n = 40), reported as mean absolute error (MAE) in days with standard deviation (SD) and standard error of the mean (SE), and 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 170) by using the specified percentage of data.

| Data% | MAE | sd | se | 95% CI |
|---|---|---|---|---|
| BrainIAC | | | | |
| 10% | 61.562375 | 60.5672891 | 9.57652926 | 19.3703588 |
| 20% | 60.945725 | 55.412646 | 8.76150863 | 17.7218239 |
| 40% | 54.1225 | 42.9808252 | 6.79586517 | 13.7459348 |
| 60% | 44.383975 | 45.4482025 | 7.18599178 | 14.5350403 |
| 80% | 44.87991 | 41.6429261 | 6.58432475 | 13.3180539 |
| 100% | 38.87375 | 37.6920367 | 5.95963427 | 12.0544981 |
| BrainSegFounder | | | | |
| 10% | 77.3231633 | 70.1168071 | 11.0864406 | 22.4244428 |
| 20% | 75.2987988 | 68.4994795 | 10.8307187 | 21.9071964 |
| 40% | 64.6278222 | 52.7894022 | 8.34673736 | 16.8828699 |
| 60% | 63.737456 | 40.2360767 | 6.36188232 | 12.8681216 |
| 80% | 63.7374917 | 39.0969842 | 6.18177599 | 12.5038222 |
| 100% | 63.737585 | 40.8706653 | 6.4622196 | 13.0710729 |

| | | | | |
|---|---|---|---|---|
| MedicalNet | | | | |
| 10% | 67.31925 | 59.4435957 | 9.39885773 | 19.0109842 |
| 20% | 71.9535 | 66.5415526 | 10.5211433 | 21.2810209 |
| 40% | 63.70275 | 57.6233011 | 9.11104389 | 18.4288257 |
| 60% | 62.82825 | 40.6724184 | 6.430874 | 13.0076704 |
| 80% | 62.403 | 42.3303829 | 6.69302121 | 13.5379132 |
| 100% | 62.241 | 40.7068276 | 6.43631458 | 13.0186751 |
| Scratch | | | | |
| 10% | 64.7765 | 55.7250351 | 8.81090169 | 17.8217308 |
| 20% | 63.28425 | 53.5982043 | 8.4746202 | 17.1415373 |
| 40% | 64.9475 | 58.5342305 | 9.25507447 | 18.7201551 |
| 60% | 59.414 | 43.1784379 | 6.82711047 | 13.8091344 |
| 80% | 63.6395 | 38.6706595 | 6.11436812 | 12.3674769 |
| 100% | 50.6305 | 42.6470037 | 6.74308335 | 13.6391735 |

Supplementary Data Table 23. Few-shot time to stroke prediction performance on the internal test set (n = 40), reported as mean absolute error (MAE) with standard deviation (SD) and standard error (SE), and 95% CI obtained from 1000 bootstrap samples. Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 2) and K = 5 (n = 10).

| K | MAE | sd | se | 95% CI |
|---|---|---|---|---|
| BrainIAC | | | | |
| K=1 | 66.105646 | 56.7890503 | 8.97913725 | 18.1620194 |
| K=5 | 69.4662829 | 65.0553664 | 10.2861566 | 20.8057156 |
| BrainSegFounder | | | | |
| K=1 | 75.9893382 | 69.3431686 | 10.9641176 | 22.1770212 |
| K=5 | 65.1539993 | 38.8729712 | 6.14635642 | 12.4321793 |
| MedicalNet | | | | |
| K=1 | 85.97616 | 74.325182 | 11.7518431 | 23.7703464 |
| K=5 | 85.158125 | 74.3898184 | 11.7620631 | 23.7910181 |
| Scratch | | | | |
| K=1 | 83.9761975 | 73.8401194 | 11.675148 | 23.6152158 |
| K=5 | 65.6385 | 39.0766309 | 6.17855784 | 12.4973128 |

Supplementary Data Table 24. MR sequence classification prediction performance on the Internal test set (n = 876) as a function of training-data proportion (10 %–100 %) evaluated using balanced accuracy (BA). All models were fully fine-tuned on the developmental cohort (n = 5004) using the varied percentage of data.

| Data% | BA |
|---|---|
| BrainIAC | |
| 10% | 0.908675799 |
| 20% | 0.945205479 |
| 40% | 0.953196347 |
| 60% | 0.97260274 |
| 80% | 0.97260274 |
| 100% | 0.971461187 |
| Scratch | |
| 10% | 0.789954338 |
| 20% | 0.848173516 |
| 40% | 0.944063927 |
| 60% | 0.953196347 |
| 80% | 0.970319635 |
| 100% | 0.96347032 |
| BrainSegFounder | |
| 10% | 0.864155251 |
| 20% | 0.929223744 |
| 40% | 0.947488584 |
| 60% | 0.955479452 |
| 80% | 0.95890411 |
| 100% | 0.960045662 |
| MedicalNet | |
| 10% | 0.742009132 |
| 20% | 0.850456621 |
| 40% | 0.896118721 |
| 60% | 0.930365297 |
| 80% | 0.950913242 |
| 100% | 0.950913242 |

Supplementary Data Table 25. Few-shot MR sequence classification prediction performance on the internal test set (n = 876), reported as balanced accuracy (BA). Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 4) and K = 5 (n = 20).

| K | BA |
|---|---|
| BrainIAC | |
| K=1 | 0.53767123 |
| K=5 | 0.73173516 |
| Scratch | |
| K=1 | 0.34246575 |
| K=5 | 0.43835616 |
| BrainSegFounder | |
| K=1 | 0.34817352 |
| K=5 | 0.40525114 |
| MedicalNet | |
| K=1 | 0.25 |
| K=5 | 0.49086758 |

Supplementary Data Table 26. Tumor segmentation performance on the internal test set (n=144) as a function of training-data proportion (10 %–100 %). Metrics include the mean Dice coefficient, standard deviation (SD), standard error of the mean (SE) and the 95 % confidence-interval margin (CI) estimated from 1,000 bootstrap samples. P-values were computed by two-sided Wilcoxon signed-rank tests comparing each model against the BrainIAC. All models were fully fine-tuned on the developmental cohort (n = 1,206) using the varied percentage of data.

| Data% | Mean Dice | SD | SE | 95% CI | P value |
|---|---|---|---|---|---|
| BrainIAC | | | | | |
| 10% | 0.72395341 | 0.15960773 | 0.01330064 | 0.02629128 | - |
| 20% | 0.76971888 | 0.13670496 | 0.01139208 | 0.02251864 | - |
| 40% | 0.77095047 | 0.13593897 | 0.01132825 | 0.02239246 | - |
| 60% | 0.78556491 | 0.12719875 | 0.0105999 | 0.02095273 | - |
| 80% | 0.78574299 | 0.13334286 | 0.01111191 | 0.02196482 | - |
| 100% | 0.79495758 | 0.12566125 | 0.01047177 | 0.02069947 | - |
| BrainSegFounder | | | | | |
| 10% | 0.7126903 | 0.16974144 | 0.01414512 | 0.02796055 | 0.60106433 |
| 20% | 0.72829382 | 0.15808672 | 0.01317389 | 0.02604073 | 0.01028084 |
| 40% | 0.6913854 | 0.18219903 | 0.01518325 | 0.03001262 | 1.07E-05 |
| 60% | 0.70938731 | 0.18831139 | 0.01569262 | 0.03101947 | 0.00011894 |

| | | | | | |
|---|---|---|---|---|---|
| 80% | 0.74125864 | 0.1670945 | 0.01392454 | 0.02752453 | 0.02361018 |
| 100% | 0.75632697 | 0.15915967 | 0.01326331 | 0.02621747 | 0.05509384 |
| MedicalNet | | | | | |
| 10% | 0.54879953 | 0.20823615 | 0.01735301 | 0.03430156 | 2.39E-15 |
| 20% | 0.54933633 | 0.20811711 | 0.01734309 | 0.03428196 | 2.06E-23 |
| 40% | 0.54881806 | 0.20785308 | 0.01732109 | 0.03423846 | 6.11E-24 |
| 60% | 0.66140363 | 0.17535723 | 0.0146131 | 0.02888561 | 4.35E-14 |
| 80% | 0.66140363 | 0.17535723 | 0.0146131 | 0.02888561 | 3.69E-14 |
| 100% | 0.6913854 | 0.18219903 | 0.01518325 | 0.03001262 | 1.07E-09 |
| Scratch | | | | | |
| 10% | 0.33591517 | 0.16859101 | 0.01404925 | 0.02777105 | 1.65E-39 |
| 20% | 0.37908154 | 0.17247161 | 0.01437263 | 0.02841027 | 1.78E-42 |
| 40% | 0.47574252 | 0.17984716 | 0.01498726 | 0.02962521 | 2.65E-36 |
| 60% | 0.66140363 | 0.17535723 | 0.0146131 | 0.02888561 | 4.35E-14 |
| 80% | 0.6913854 | 0.18219903 | 0.01518325 | 0.03001262 | 3.40E-08 |
| 100% | 0.70938731 | 0.18831139 | 0.01569262 | 0.03101947 | 4.29E-06 |

Supplementary Data Table 27. Tumor segmentation performance on the internal test set (n=144). Metrics include the mean Dice coefficient, standard deviation (SD), standard error of the mean (SE) and the 95 % confidence-interval margin (CI) estimated from 1,000 bootstrap samples. P-values were computed by two-sided Wilcoxon signed-rank tests comparing each model against the BrainIAC. Models were trained on the developmental cohort under two few-shot configurations: K = 1 (n = 2) and K = 5 (n = 10).

| K | Mean Dice | SD | SE | 95% CI | P-value |
|---|---|---|---|---|---|
| BrainIAC | | | | | |
| K=1 | 0.51847558 | 0.23296239 | 0.01941353 | 0.03837458 | - |
| K=5 | 0.61565684 | 0.188861 | 0.01573842 | 0.03111 | - |
| BrainSegFounder | | | | | |
| K=1 | 0.49987587 | 0.21099174 | 0.01758264 | 0.03475548 | 0.35583218 |
| K=5 | 0.53735578 | 0.21261839 | 0.0177182 | 0.03502343 | 0.00379721 |
| MedicalNet | | | | | |
| K=1 | 0.13560138 | 0.112372 | 0.00936433 | 0.01851041 | 1.81E-34 |
| K=5 | 0.51847558 | 0.23296239 | 0.01941353 | 0.03837458 | 8.24E-04 |
| Scratch | | | | | |
| K=1 | 0.03501 | 0.10676127 | 0.00889677 | 0.01758618 | 7.30E-45 |

| | | | | | |
|---|---|---|---|---|---|
| K=5 | 0.13651246 | 0.11221316 | 0.0093511 | 0.01848424 | 1.03E-42 |

Supplementary Data Table 28. Linear-probe brain age prediction performance on the external test set (n = 1,072), shown as mean absolute error (MAE) with 95 % confidence intervals (CI) obtained from 1,000 bootstrap samples. Models were trained on the full developmental cohort (n = 3,882) by freezing the backbone and fitting only the regression head.

| Approach | MAE | 95% CI |
|---|---|---|
| BrainIAC | 7.51 | 0.21 |
| BrainSegFounder | 10.45 | 0.23 |
| MedicalNet | 7.9 | 0.2 |

Supplementary Data Table 29. Linear-probe IDH mutation prediction performance on the internal test set (n = 99), reported as AUC with 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 396) by freezing the backbone and fitting only the classification head.

| Approach | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | 0.68758573 | 0.54525034 | 0.82512003 |
| BrainSegFounder | 0.60562414 | 0.46774691 | 0.72913237 |
| MedicalNet | 0.58161866 | 0.43209877 | 0.71550069 |

Supplementary Data Table 30. Linear-probe One-year overall survival (binary classification) prediction performance on the internal test set (n = 134), reported as AUC with 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 545) by freezing the backbone and fitting only the classification head.

| Approach | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | 0.64292115 | 0.55263217 | 0.7327677 |
| BrainSegFounder | 0.53259409 | 0.44016017 | 0.63419579 |
| MedicalNet | 0.54177867 | 0.43546147 | 0.63508065 |

Supplementary Data Table 31. Linear-probe Mild Cognitive Impairment (MCI) prediction performance on the internal test set (n = 40), reported as AUC with 95% CI obtained from 1000 bootstrap samples. Models were trained on the full developmental cohort (n = 195) by freezing the backbone and fitting only the classification head.

| Approach | AUC | 95% CI low | 95% CI high |
|---|---|---|---|
| BrainIAC | 0.72100313 | 0.53597179 | 0.87782132 |
| BrainSegFounder | 0.47335423 | 0.28213166 | 0.69286834 |
| MedicalNet | 0.47021944 | 0.27899687 | 0.68338558 |

Supplementary Data Table 32. Linear-probe time to stroke prediction performance on the test set (n = 40), shown as mean absolute error (MAE) with 95 % confidence intervals (CI) obtained from 1,000 bootstrap samples. Models were trained on the full developmental cohort (n = 170) by freezing the backbone and fitting only the regression head.

| Approach | MAE | 95% CI |
|---|---|---|
| BrainIAC | 64.31 | 16.35 |
| BrainSegFounder | 72.6 | 21.13 |
| MedicalNet | 74.7 | 21.76 |

Supplementary Data Table 33. Linear-probe MR sequence classification prediction performance on the internal test set (n = 876), reported as balanced accuracy (BA). Models were trained on the full developmental cohort (n = 5,004) by freezing the backbone and fitting only the classification head.

| Approach | BA |
|---|---|
| BrainIAC | 0.75799087 |
| BrainSegFounder | 0.42465753 |
| MedicalNet | 0.40182648 |

Supplementary Data Table 34. Linear-probe Tumor segmentation performance on the internal test set (n=144). Metrics include the mean Dice coefficient, standard deviation (SD), standard error of the mean (SE) and the 95 % confidence-interval margin (CI) estimated from 1,000

bootstrap samples. P-values were computed by two-sided Wilcoxon signed-rank tests comparing each model against the BrainIAC. Models were trained on the full developmental cohort (n = 1,206) by freezing the backbone encoder and fitting only the decoder head.

| Approach | Mean Dice | SD | SE | 95% CI | P-value |
|---|---|---|---|---|---|
| BrainIAC | 0.67087045 | 0.15526345 | 0.01293862 | 0.02557567 | |
| BrainSegFounder | 0.64565693 | 0.17087852 | 0.01423988 | 0.02814785 | 0.24047353 |
| MedicalNet | 0.57807271 | 0.1786579 | 0.01488816 | 0.02942931 | 1.69E-06 |

Supplementary Data Table 35. SimCLR pretraining hyperparameters. 3D ViT-B was chosen as the convolutional backbone, the pretraining was done on 1* 48GB Nvidia A6000 GPU.

| Hyper-Parameter | Value |
|---|---|
| Model | |
| Convolutional layers | 50 |
| FC layers | 2 |
| Parameters (million) | 52.5 |
| Latent dimension | 1024 |
| Model size (mb) | 209.8 |
| Input shape (H,W,D) | 128,128,128 |
| Transforms | |
| Crop scale (min,max) | 0.2,1 |
| Resizing mode | trilinear |
| Rotate range (min, max) (radian) | -0.34 - 0.34 |
| Translate range (pixels) | -15 - 15 |
| Scale range | 0.0 – 1.0 |
| Flip axis | 0,1,2 |
| Gaussian smooth range (sigma) | 0.25 – 1.5 |
| Shift intensity range | 50 – 100 |
| Contrast range (gamma) | 0.5 – 2.0 |
| Gaussian smooth range (std) | 0.0 – 0.09 |
| Training | |
| Loss | NTXent |
| Scheduler | Cosine |

| | |
|---|---|
| Learning Rate | 0.0001 |
| Momentum | 0.9 |
| Decay | 0.00005 |
| Optimizer | SGD |

## A.1 Primary Datasets

### ABCD

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development SM (ABCD) Study (https://abcdstudy.org), held in the NIMH Data Archive (NDA). This is a multisite, longitudinal study designed to recruit more than 10,000 children age 9-10 and follow them over 10 years into early adulthood. The ABCD Study® is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, U24DA041147. A full list of supporters is available at https://abcdstudy.org/federal-partners.html. A listing of participating sites and a complete listing of the study investigators can be found at https://abcdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the authors' views and may not reflect the opinions or views of the NIH or ABCD consortium investigators. The ABCD data repository grows and changes over time. The ABCD data used in this report came from the fast-track data release. The raw data are available at https://nda.nih.gov/edit_collection.html?id=2573. Instructions on how to create an NDA study are available at https://nda.nih.gov/training/modules/study.html). Additional support for this work was made possible from supplements to U24DA041123 and U24DA041147, the National Science Foundation (NSF 2028680), and Children and Screens: Institute of Digital Media and Child Development Inc. (Casey et al., 2018)

### ADNI

Data used in the preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer’s disease (AD).

Data collection and sharing for this project was funded by the Alzheimer's Disease Neuroimaging Initiative (ADNI) (National Institutes of Health Grant U01 AG024904) and DOD ADNI (Department of Defense award number W81XWH-12-2-0012). ADNI is funded by the National Institute on Aging, the National Institute of Biomedical Imaging and Bioengineering, and through generous contributions from the following: AbbVie, Alzheimer’s Association; Alzheimer’s Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers

Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC.; Johnson & Johnson Pharmaceutical Research & Development LLC.; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC.; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics. The Canadian Institutes of Health Research is providing funds to support ADNI clinical sites in Canada. Private sector contributions are facilitated by the Foundation for the National Institutes of Health (www.fnih.org). The grantee organization is the Northern California Institute for Research and Education, and the study is coordinated by the Alzheimer's Therapeutic Research Institute at the University of Southern California. ADNI data are disseminated by the Laboratory for Neuro Imaging at the University of Southern California.

**DFCI/BCH LGG**

The DFCI/BCH brain tumor dataset contains private hospital data that is controlled due to privacy concerns. Access to the derived dataset will be considered upon request to the corresponding author (Benjamin H. Kann, M.D., email: Benjamin_Kann@dfci.harvard.edu, timeframe for response 2 weeks).

**OASIS-3**

Data were provided by OASIS , OASIS-3: Longitudinal Multimodal Neuroimaging: Principal Investigators: T. Benzinger, D. Marcus, J. Morris; NIH P30 AG066444, P50 AG00561, P30 NS09857781, P01 AG026276, P01 AG003991, R01 AG043434, UL1 TR000448, R01 EB009352. AV-45 doses were provided by Avid Radiopharmaceuticals, a wholly owned subsidiary of Eli Lilly. OASIS-3_AV1451: Principal Investigators: T. Benzinger, J. Morris; NIH P30 AG066444, AW00006993. AV-1451 doses were provided by Avid Radiopharmaceuticals, a wholly owned subsidiary of Eli Lilly.

**MCSA**

The data contained in this analysis were obtained under research grant U01 AG006786 (PI: Ronald C. Petersen, M.D., Ph. D.) from the National Institutes of Health to the Mayo Clinic Study of Aging

**ABIDE**

Primary support for the work by Adriana Di Martino was provided by the (NIMH K23MH087770) and the Leon Levy Foundation.
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**CBTN LGG**

The CBTN data is available upon request at " https://cbtn.org/ "

**MIRIAD**

Data used in the preparation of this article were obtained from the MIRIAD database. The MIRIAD investigators did not participate in analysis or writing of this report. The MIRIAD dataset is made available through the support of the UK Alzheimer's Society (Grant RF116). The

original data collection was funded through an unrestricted educational grant from GlaxoSmithKline (Grant 6GKC)."

**PPMI**

Data used in the preparation of this article was obtained on [2024-08-15] from the Parkinson's Progression Markers Initiative (PPMI) database (www.ppmi-info.org/access-dataspecimens/download-data), RRID:SCR_006431. For up-to-date information on the study, visit www.ppmi-info.org. PPMI – a public-private partnership – is funded by the Michael J. Fox Foundation for Parkinson's Research and funding partners, including 4D Pharma, Abbvie, AcureX, Allergan, Amathus Therapeutics, Aligning Science Across Parkinson's, AskBio, Avid Radiopharmaceuticals, BIAL, BioArctic, Biogen, Biohaven, BioLegend, BlueRock Therapeutics, Bristol-Myers Squibb, Calico Labs, Capsida Biotherapeutics, Celgene, Cerevel Therapeutics, Coave Therapeutics, DaCapo Brainscience, Denali, Edmond J. Safra Foundation, Eli Lilly, Gain Therapeutics, GE HealthCare, Genentech, GSK, Golub Capital, Handl Therapeutics, Insitro, Jazz Pharmaceuticals, Johnson & Johnson Innovative Medicine, Lundbeck, Merck, Meso Scale Discovery, Mission Therapeutics, Neurocrine Biosciences, Neuron23, Neuropore, Pfizer, Piramal, Prevail Therapeutics, Roche, Sanofi, Servier, Sun Pharma Advanced Research Company, Takeda, Teva, UCB, Vanqua Bio, Verily, Voyager v. 25MAR2024 Therapeutics, the Weston Family Foundation and Yumanity Therapeutics.

**OASIS-2**

OASIS-2: Longitudinal: Principal Investigators: D. Marcus, R, Buckner, J. Csernansky, J. Morris; P50 AG05681, P01 AG03991, P01 AG026276, R01 AG021910, P20 MH071616, U24 RR021382

**DFCI/BCH HGG**

The DFCI/BCH brain tumor dataset contains private hospital data that is controlled due to privacy concerns. Access to the derived dataset will be considered upon request to the corresponding author (Benjamin H. Kann, M.D., email: Benjamin_Kann@dfci.harvard.edu, timeframe for response 2 weeks).

**UPENN-GBM**

Bakas, S., Sako, C., Akbari, H., Bilello, M., Sotiras, A., Shukla, G., Rudie, J. D., Flores Santamaria, N., Fathi Kazerooni, A., Pati, S., Rathore, S., Mamourian, E., Ha, S. M., Parker, W., Doshi, J., Baid, U., Bergman, M., Binder, Z. A., Verma, R., … Davatzikos, C. (2021). Multi-parametric magnetic resonance imaging (mpMRI) scans for de novo Glioblastoma (GBM) patients from the University of Pennsylvania Health System (UPENN-GBM) (Version 2) [Data set]. The Cancer Imaging Archive. https://doi.org/10.7937/TCIA.709X-DN49

**Wu1200**

This HCP data release includes high-resolution 3T MR scans from young healthy adult twins and non-twin siblings (ages 22-35) using four imaging modalities: structural images (T1w and T2w), resting-state fMRI (rfMRI), task-fMRI (tfMRI), and high angular resolution diffusion imaging (dMRI). Behavioral and other individual subject measure data (both NIH Toolbox and non-Toolbox measures) is available on all subjects. MEG data and 7T MR data is available for a subset of subjects (twin pairs). The Open Access Dataset includes imaging data and most behavioral data. All details in the imaging protocols can be found at study webpage (https://humanconnectome.org/study/hcpyoung-adult/document/1200-subjects-data-release/)

**LONG579**

The public neuroimaging and behavioral dataset entitled “A longitudinal neuroimaging dataset on language processing in children ages 5, 7, and 9 years old” available on the OpenNeuro project (https://openneuro.org) and organized in compliance with the Brain Imaging Data Structure (BIDS). It includes 322 participants, recruited from the Austin, Texas. All neuroimaging data were collected using a Siemens Skyra 3 T MRI scanner located at The University of Texas at Austin Imaging Research Center. All images were acquired using a 64-channel head coil. Participants were positioned supine in the MRI scanner and foam pads were placed around the head to minimize movement. T1- weighted Magnetization Prepared - RApid Gradient Echo (MPRAGE) images were collected using GRAPPA, a parallel imaging technique based on k-space, and the following parameters: GRAPPA accel.factor PE = 2, TR = 1900 ms, TE = 2.43 ms, field of view = 256 mm, matrix size = 256 × 256, bandwidth = 180 Hz/Px, slice thickness = 1 mm, number of slices = 192, voxel size = 1 mm isotropic, flip angle = 9°. (Wang et al., 2022)

**BABY**

The Baby Connectome Project (BCP: https://nda.nih.gov/edit_collection.html?id=2848) is a four-year study of children from birth through five years of age, intended to provide a better understanding of how the brain develops from infancy through early childhood and the factors that contribute to healthy brain development. This project is a research initiative of the Neuroscience Blueprint – a cooperative effort among the 15 NIH Institutes, Centers, and Offices that support neuroscience research. The BCP is supported by Wyeth Nutrition through a donation to the FNIH. Images are acquired on 3T Siemens Prisma MRI scanners using a Siemens 32-channel head coil at the Center for Magnetic Resonance Research (CMRR) at the University of Minnesota and the Biomedical Research Imaging Center (BRIC) at the University of North Carolina at Chapel Hill (Howell et al., 2019).

**AOMIC**

The Amsterdam Open MRI Collection (AOMIC, https://openneuro.org/datasets/ds003097/versions/1.2.1) is a collection of three datasets with multimodal (3T) MRI data, including structural (T1-weighted), diffusionweighted, and (resting-state and task-based) functional BOLD MRI data, as well as detailed demographics and psychometric variables from a large set of healthy participants (N = 928, N = 226, and N = 216). Data from all three datasets were acquired on the same Philips 3T scanner (Philips, Best, the Netherlands) but underwent several upgrades in between the three studies (Snoek et al., 2021).

**Calgary**

The Preschool MRI study in The Developmental Neuroimaging Lab at the University of Calgary uses different magnetic resonance imaging (MRI) techniques to study brain structure and function in early childhood (https://osf.io/axz5r/files/osfstorage). All imaging for this dataset was conducted using the same General Electric 3T MR750w system and 32-channel head coil (GE, Waukesha, WI) at the Alberta Children’s Hospital in Calgary, Canada. Children were scanned either while awake and watching a movie, or while sleeping without sedation. The University of Cal- gary Conjoint Health Research Ethics Board (CHREB) approved this study (REB13-0020). T1-weighted images were acquired using an FSPGR BRAVO sequence with TR = 8.23 ms, TE = 3.76 ms, TI = 540 ms, flip angle=12 degrees, voxel size = 0.9x0.9x0.9 mm3, 210 slices, matrix size=512x512, field of view=23.0 cm. ASL images were acquired with the vendor supplied pseudo continuous 3D ASL sequence with TR = 4.56 s, TE = 10.7 ms, in-plane reso- lution of 3.5x3.5 mm2, post label delay of 1.5 s, and thirty 4.0 mm thick slices. The sequence scan time was 4.4 minutes (Reynolds et al., 2020)

**NIMH**

The data used in this work was collected from the 5.1 release (https://nda.nih.gov/edit_collection.html?id=1151) . MRI scans were acquired using either General Electric or Siemens 1.5 Tesla scanners involving six sites or Pediatric Study Centers (PSC) in the United States. The Institutional Review Board at the University of Wisconsin-Madison also approved the analysis of the data of this human subject. Sequence type: 3D FLASH/SPGR; GE sequence: pulse sequence=SPGR, mode=3D; TR: 22 ms; TE: 10-11 ms; excitation pulse angle: 30 degrees; orientation: sagittal; FoV: 250mmISx250mmAP; matrix: 256 x 256 ( x 124 - 180 slices); slices: 160- 180 slices of 1-1.5 mm thickness (cover entire head). Note that on GE systems with a 124-slice limitation, slice thickness should be adjusted to cover the entire head with 124 slices: signal averages: 1; scan time: 11.6 – 16.8 min (Evans, 2006).

**ICBM**

Data used in the preparation of this work were obtained from the International Consortium for Brain Mapping (ICBM) database (www.loni.usc.edu/ICBM). The ICBM project (Principal Investigator John Mazziotta, M.D., University of California, Los Angeles) is supported by the National Institute of Biomedical Imaging and BioEngineering. ICBM is the result of efforts of co-investigators from UCLA, Montreal Neurologic Institute, University of Texas at San Antonio, and the Institute of Medicine, Juelich/Heinrich Heine University - Germany. Data collection and sharing for this project was provided by the International Consortium for Brain Mapping (ICBM; Principal Investigator: John Mazziotta, MD, PhD). ICBM funding was provided by the National Institute of Biomedical Imaging and BioEngineering. ICBM data are disseminated by the Laboratory of Neuro Imaging at the University of Southern California (Kötter et al., 2001).

**IXI**

The data has been collected at three different hospitals in London:Hammersmith Hospital using a Philips 3T system (details of scanner parameters: http://braindevelopment.org/scanner-philips-medical-systems-intera-3t/), Guy's Hospital using a Philips 1.5T system (details of scanner parameters: http://braindevelopment.org/scanner-philips-medical-systems-gyroscan-intera-1-5t/), Institute of Psychiatry using a GE 1.5T system (details of the scan parameters not available at the moment). The Thames Valley MREC granted ethical approval. The T1 and T2 images were acquired prior to diffusion-weighted imaging using 3D MRPRAGE and dual-echo weighted imaging (IXI Dataset – Brain Development, n.d.).

**NYU2(CoRR)**

The Consortium for Reliability and Reproducibility (CoRR, http://fcon_1000.projects.nitrc.org/fcpClassic/FcpTable.html ) has aggregated 1,629 typical individuals' resting state fMRI (rfMRI) data (5,093 rfMRI scans) from 18 international sites and is openly sharing them via the International Data-sharing Neuroimaging Initiative (INDI). In this study, we used a subset from CoRR study "NYU 2" created by New York University (Di Martino, Kelly)(An Open Science Resource for Establishing Reliability and Reproducibility in Functional Connectomics | Scientific Data, n.d.).

**PING**

The PING Data Resource(https://nda.nih.gov/edit_collection.html?id=2607) is the product of a multi-site project involving developmental researchers across the United States, including UC San Diego, the University of Hawaii UC Los Angeles Children's Hospital of Los Angeles of the University of Southern California UC Davis Kennedy Krieger Institute of Johns Hopkins University Sackler Institute of Cornell University University of Massachusetts Massachusetts

General Hospital at Harvard University and Yale University. The Data Resource includes neurodevelopmental histories, information about developing mental and emotional functions, multimodal brain imaging data, and genotypes for well over 1000 children and adolescents between the ages of 3 and 20. The PING imaging protocol takes advantage of key technologies developed for the consortium and builds on earlier methods development performed as part of the Biomedical Informatics Research Network (BIRN (Keator et al., 2008) and the Alzheimer's Disease Neuroimaging Initiative (ADNI (The Alzheimer's Disease Neuroimaging Initiative (ADNI): MRI Methods - Jack - 2008 - Journal of Magnetic Resonance Imaging - Wiley Online Library, n.d.)). Specifically, a standard PING scan session included: 1) a 3D T1-weighted inversion prepared RF-spoiled gradient echo scan using prospective motion correction (PROMO), for cortical and subcortical segmentation; 2) a 3D T2-weighted variable flip angle fast spin echo scan, also using PROMO, for detection and quantification of white matter lesions and segmentation of VV; 3) a high angular resolution diffusion imaging (HARDI) scan, with integrated B0 distortion correction (DISCO), for segmentation of white matter tracts and measurement of diffusion parameters; and 4) a resting state blood oxygenation leveldependent (BOLD) fMRI scan, with integrated distortion correction. Pulse sequence parameters used across (3 T) scanner manufacturers (GE, Siemens, and Phillips) and models were optimized for equivalence in contrast properties and consistency in imagederived quantitative measures (Jernigan et al., 2016).

## Pixar

One hundred twenty-two 3.5–12-year-old children (M(s.d.) = 6.7(2.3); 64 females) participated in the study (https://openfmri.org/dataset/ds000228/). Child and adult participants were recruited from the local community. All adult participants gave written consent; parent/guardian consent and child assent was received for all child participants. Recruitment and experiment protocols were approved by the Committee on the Use of Humans as Experimental Subjects (COUHES) at the Massachusetts Institute of Technology. Whole-brain structural and functional MRI data were acquired on a 3- Tesla Siemens Tim Trio scanner located at the Athinoula A. Martinos Imaging Center at MIT. Children under age 5 years used one of two custom 32-channel phased-array head coils made for younger (n = 3, M(s.d.) = 3.91(.42) years) or older (n = 28, M(s.d.) = 4.07(.42) years) children; all other participants used the standard Siemens 32-channel head coil. T1-weighted structural images were collected in 176 interleaved sagittal slices with 1 mm isotropic voxels (GRAPPA parallel imaging, acceleration factor of 3; adult coil: FOV: 256 mm; kid coils: FOV: 192 mm). Functional data were collected with a gradient-echo EPI sequence sensitive to Blood Oxygen Level Dependent (BOLD) contrast in 32 interleaved near-axial slices aligned with the anterior/posterior commissure and covering the whole brain (EPI factor: 64; TR: 2 s, TE: 30 ms, flip angle: 90°). This data was obtained from the OpenfMRI database, accession number is ds000228. Dataset version 1.0.2 (MRI Data of 3-12 Year Old Children and Adults during Viewing of a Short Animated Film, n.d.)

## SALD

The data was generated in the Southwest University Adult Lifespan Dataset (SALD), which comprises a large cross-sectional sample (n = 494; age range = 19-80) undergoing a multi-modal (sMRI, rs-fMRI, and behavioral). All data were collected at the Southwest University Center for Brain Imaging using a 3.0-T Siemens Trio MRI scanner (Siemens Medical, Erlangen, Ger- many). A magnetization-prepared rapid gradient echo (MPRAGE) sequence was used to acquire high-resolution T1-weighted anatomical images (repetition time=1,900ms, echo time=2.52ms, inversion time=900ms, flip angle=90 degrees, resolution matrix=256×256, slices=176, thickness =1.0mm, and voxel size=111mm3) (Wei et al., 2018).

**OASIS-1**

OASIS-1: Cross-Sectional: Principal Investigators: D. Marcus, R, Buckner, J, Csernansky J. Morris; P50 AG05681, P01 AG03991, P01 AG026276, R01 AG021910, P20 MH071616, U24 RR021382

**ATLAS**

The archiving of this dataset was specifically supported by the NIH-funded Center for Large Data Research and Data Sharing in Rehabilitation (CLDR) under a Category 2 Pilot Grant (P2CHD06570) and NIH R01 NS115845 and NIH K01 HD091283.

## A.2 Downstream Evaluation

### A.2.1 Sequence classification adaptation

For the sequence classification application, we formulated a four-way classification task to distinguish between T1, T2, FLAIR, and T1CE MR imaging sequences. The architecture comprises of vision encoder generating 2048-dimensional latent features, followed by a fully connected layer of matching dimensionality, followed by a four-neuron output layer. Training for all three approaches was implemented using cross-entropy loss under supervised conditions.

The model was optimized using Adam optimizer with approach-specific learning rates: 0.001 for scratch initialization and 0.0001 for both BrainIAC, BrainSegFounder, and MedicalNet approaches. Training was performed for 100 epochs with a batch size of 16, with a CrossEntropyLoss and a Reduce-LR-On-Plateau learning rate scheduler. Model selection was based on validation set performance, with the best-performing checkpoints from each strategy retained for holdout set evaluation.

We utilized the BraTS23 dataset, comprising 5,880 scans equally distributed across the four sequence classes. We quantified model performance using balanced accuracy (BA) as the primary metric. To evaluate the subgroup performance, we performed pairwise comparisons between sequences across all initialization strategies, using BA to quantify performance differences (Extended Data Figure 3).

### A.2.2 Brain age prediction

For the Brain age application, we formulated a regression task with Brain age being the regression variable. The pipeline consists of single input T1 image passed into a vision encoder followed by a fully connected layer of 2048 dimensions, which is further connected to a single neuron output for regression. The training is done in a supervised fashion with mean absolute error (MAE) as the primary loss function employed. We also use MAE as the performance evaluation metric. The individual data fraction models were trained using a batch size 16, with initial learning rate of 0.001(scratch) and 0.0001 (BrainIAC , BrainSegFounder, and MedicalNet) with a MSE loss, Adam optimizer and ReduceLROnPlateau learning rate scheduler for 100 epochs, and mean absolute error (MAE) as the evaluation metric.

We aggregated a dataset pool of 14 datasets where we separated the dataset pool into a split of in-distribution (development) and out of distribution set. The in-distribution dataset consisted of (10) and out of distribution dataset (1072 images) consisted of (4) (Supplementary data Table 7). The in-distribution dataset was further pooled and split into a training-validation (3882 images) and holdout-testing split (1295 images) (with 80:20 ratio), where the training-validation split was further fragmented into dataset% fractions (Extended Data Figure 1). All the dataset fraction models were tested on in-distribution and out-of-distribution test sets.

### A.2.3 Mutation classification

For the mutation classification application, we formulated the problem as a binary classification task for IDH mutation (IDH mutated Vs Wildtype) for diffuse glioma subjects from the UCSF-PDGM dataset containing 392 IDH mutated and 103 wildtype subjects. The pipeline, taking in two images (T1CE, FLAIR) from same subject as an input, consisted of the vision encoder which outputted two feature vectors (2048 dimensions) corresponding to the two input images. This was followed by the average pooling layers which results in a single feature vector of dimension 2048 that is passed into the fully connected layer with 2048 neurons and finally the output layer consisted of a single neuron for the binary classification. The training procedure consisted of binary cross entropy loss (BCE) as the optimization metric along with Adam optimizer with the learning rate of 0.001 (Scratch) and 0.0001 (MedicalNet, BrainSegFounder ,and BrainIAC) along with ReduceLROnPlateau scheduler. We used AUC as the performance evaluation and comparison metric along with balanced accuracy.

### A2.4 Overall survival prediction

For overall survival prediction, we developed a classification pipeline to predict one-year overall survival outcomes in glioblastoma patients using the UPENN-GBM dataset, with 668 subjects—360 of whom survived beyond one year. We reserved 534 subjects for training-validation and 134 as a holdout test set. We curated another external test set from TCGA-GBM/BraTS23 dataset of 134 subjects with complete survival information. We used 1-year overall survival as the primary endpoint. The pipeline processed four MRI modalities per subject: T1CE, FLAIR, T1, and T2 scans. Each image is input into the vision encoder, generating four separate 2048-dimensional feature vectors which were pooled using average pooling layer generating a single 2048-dimensional feature vector. This unified vector was then passed through a fully connected layer with 2048 neurons, followed by an output layer of a single neuron for classification.
We computed a risk score for each patient based on the model's output, using the median risk score from the training set as the threshold to stratify patients into high-risk and low-risk groups. Kaplan-Meier survival analysis and log-rank tests were conducted to evaluate the effectiveness of this stratification.
The pipeline was trained in a supervised fashion using a BCE loss and Adam optimizer. Learning rates were set at 0.001 (Scratch) and 0.0001 (MedicalNet, BrainSegFounder , and BrainIAC), along with a ReduceLROnPlateau scheduler over 200 epochs. The model

performance was evaluated using AUC and Concordance Index (CI) metric. AUC was reported for individuals at risk up to the one-year mark (i.e., not censored and no event); five patients in our external test data met this criterion and were excluded from AUC calculation (Supplementary Data Table 16).

### A2.5 MCI classification

For mild cognitive impairment (MCI) classification, we developed a binary classification framework using the OASIS-1 dataset to distinguish between MCI subjects (label 1) and healthy controls (label 0) on total of 235 scans (1 T1w scan per subject) by selecting the first scan for each subject. Subjects were stratified based on their Clinical Dementia Rating (CDR), with scores of 0.5, 1.0, and 2.0 categorized as MCI (label 1). The final cohort comprised 100 MCI subjects and 135 healthy controls. The pipeline, inputting T1w image, consisted of a vision encoder (latent dimension of 2048) followed by the fully connected (FC) and a single output neuron for the binary classification. The training procedure consisted of binary cross entropy loss (BCE) as the optimization metric along with Adam optimizer and ReduceLROnPlateau scheduler. We used AUC as the performance evaluation and comparison metric.

### A2.6 Time-to-stroke prediction

For the time-to-stroke prediction, we formulated a regression task with time since stroke onset being the regression variable from the ATLAS dataset. The study cohort consisted of 210 subjects with median time to stroke 77 days [IQR: 11.0-151.0]. The pipeline consisted of single input T1 image passed into the vision encoder followed by a fully connected layer of 2048 dimensions and a single neuron output for regression.

The training is done in a supervised fashion with mean a combination of MSE, MAE and Huber Loss as the primary loss function employed (Equation 1). We used MAE as the performance evaluation metric.

$$\alpha \times \mathrm{MSE} + (1 - \alpha) \times (0.5 \times \mathrm{MAE} + 0.5 \times \mathrm{Huber}) \qquad (1)$$

### A2.7 Tumor Segmentation adaptation

For the tumor segmentation task, we utilized the publicly available training split of the BraTS23 adult glioma dataset, further dividing it into training (N = 936), validation (N = 270), and testing (N = 144) subsets. The provided multi-class tumor annotations — Label 1 (NCR/NET), Label 2 (ED), and Label 3 (ET) — were combined into a single binary mask. FLAIR sequences were used as the input images, defining the task as a single-image binary segmentation problem. The downstream segmentation models were constructed by adapting each pre-trained backbone accordingly: for the 3D ResNet50 (SimCLR), an upsampling decoder was appended after the “Layer4” block; for the 3D ViT-B (SimCLR) encoder, we integrated it into a UNETR architecture; and for the SwinViT (MAE) encoder, we employed a SwinUNETR configuration. Additionally, a UNETR initialized with random weights was trained from scratch to serve as a

baseline for downstream performance evaluation. All models were trained using a combination of Dice loss and binary cross-entropy loss, optimized with the AdamW optimizer and a cosine annealing learning rate scheduler. Data augmentations included random flipping, rotation, 3D elastic deformations, bias field, and Gaussian noise to improve generalizability.

### A. 3 Few-shot Analysis

Few-shot adaptation capabilities were implemented in two experimental settings: single-sample (K=1) and five-sample (K=5) per class (Training and validation), with entire model finetuned end-to-end and performance assessed on independent holdout test sets across all downstream tasks. For sequence classification with four modality classes (T1W, T2W, FLAIR, T1ce), the training and validation split consisted of 4 samples for K=1 and 20 samples for K=5 each. Brain age prediction, with four age-based bins, similarly used 4 samples for K=1 and 20 samples for K=5. For binary classification tasks (IDH mutation, overall survival, and MCI classification), we employed 2 samples for K=1 and 10 samples for K=5. For time-to-stroke prediction task, we stratified time since stroke onset into two temporal bins (0-100, 100 and above) and utilized 2 samples for K=1 and 10 samples for K=5. Supervised fine-tuning was performed consistently across all the approaches (BrainIAC, BrainSegFounder , MedicalNet, and Scratch) with batch sizes of 2 and 10 for K=1 and K=5 settings, respectively.

### A.4 Model Stability Analysis

To determine the stability and resiliency of different models to imaging artifacts and common perturbations seen across MRI scanners, we synthetically simulated image perturbations in the form of contrast shifts, Gibbs artifacts, and bias fields, and evaluated the performance of BrainIAC, MedicalNet, BrainSegFounder, and Scratch models across six downstream applications (MRI sequence classification, brain age prediction, tumor IDH mutation prediction, overall survival in GBM, MCI classification, and time-to-stroke prediction) under varying levels of three imaging parameters: image contrast (scale: 0.5–2.0), Gibbs artifact (scale: 0.0–0.4), and bias field (scale: 0.0–0.4) at 100% data availability for fine-tuning. The contrast perturbation was applied using the AdjustContrast function from the MONAI framework, with a gamma scale ranging from -0.5 to 2, where 1 represents the normal condition. Gibbs perturbation was introduced using the GibbsNoise function from MONAI, with an alpha scale ranging from 0.0 to 0.4, and 0.0 representing the normal condition. Bias perturbation was performed using the RandBiasField function from MONAI, with a bias scale ranging from 0.0 to 0.4, where 0.0 denotes the normal state.

### A.5 Saliency Map and dimensionality reduction

To identify the regions in the images where the models focus most—leading to activation of model weights—we selected random images from the holdout set of each downstream application. We generated saliency maps for both the foundation model (BrainIAC) and the fine-

tuned BrainIAC models adapted for each downstream task (Figure 7). We employed the smooth guided backpropagation approach to produce the saliency heatmaps, which involved stopping the negative gradients and allowing only activated neurons during backpropagation. This method is implemented in the MONAI framework as GuidedBackpropSmoothGrad.

We performed dimensionality reduction using t-distributed stochastic neighbor embedding (t-SNE) to visually represent the high-dimensional latent features (2048 dimensions). The features were reduced to two dimensions, and each point on the t-SNE map was color-coded based on age group binning for the Brain age task (Figure 6b,d).